Optimization: Critical Points & Lagrange Multipliers

ANSWER KEY

Section 1: Finding Critical Points

1 What condition defines a critical point (x_0, y_0) of a differentiable function $f(x, y)$?

a. $\nabla f(x_0, y_0) = \mathbf{0}$**, meaning both** $f_x(x_0, y_0) = 0$ **and** $f_y(x_0, y_0) = 0$ ✓

b. $f_{xx}(x_0, y_0) = 0$

c. $f(x_0, y_0) = 0$

d. $f_x(x_0, y_0) = f_y(x_0, y_0)$

Critical points occur where the gradient vector is zero (both first-order partials vanish simultaneously) or where a partial derivative fails to exist.

2 Find the critical points of $f(x, y) = x^2 + y^2 - 4x + 6y + 10$.

a. $(2, -3)$ ✓

b. $(-2, 3)$

c. $(4, -6)$

d. $(0, 0)$

$f_x = 2x - 4 = 0 \Rightarrow x = 2$. $f_y = 2y + 6 = 0 \Rightarrow y = -3$. The only critical point is $(2, -3)$.

3 Find all critical points of $g(x, y) = x^3 - 3x + y^2$.

a. $(1, 0)$ **and** $(-1, 0)$ ✓

b. $(1, 0)$ only

c. $(0, 0)$

d. $(3, 0)$

$g_x = 3x^2 - 3 = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$. $g_y = 2y = 0 \Rightarrow y = 0$. The critical points are $(1, 0)$ and $(-1, 0)$.

Section 2: The Second Partials Test (Hessian Discriminant)

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4 What is the definition of the Hessian discriminant D in the Second Partials Test?

a. $D = f_{xx}f_{yy} - (f_{xy})^2$ ✓

b. $D = f_{xx} + f_{yy}$

c. $D = (f_{xy})^2 - f_{xx}f_{yy}$

d. $D = f_{xx}f_{yy} - f_{xy}$

The discriminant D is the determinant of the Hessian matrix $[[f_{xx}, f_{xy}], [f_{yx}, f_{yy}]]$, which is $D = f_{xx}f_{yy} - (f_{xy})^2$.

5 If at a critical point $D > 0$ and $f_{xx} > 0$, what does the test conclude?

a. f **has a local minimum at the point** ✓

b. f has a local maximum at the point

c. The point is a saddle point

d. The test is inconclusive

When $D > 0$, the surface is locally bowl-shaped; if $f_{xx} > 0$, it curves upward in all directions, establishing a local minimum.

6 If at a critical point $D < 0$, what does the test conclude?

a. f **has a saddle point** ✓

b. f has a local maximum

c. f has a local minimum

d. The test is inconclusive

When $D < 0$, the surface curves upward in one direction and downward in another, producing a saddle point.

Section 3: Classifying Extrema and Inconclusive Cases

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7 Classify the critical point $(0, 0)$ for $f(x, y) = x^2 - y^2$.

a. **Saddle point ✓**

b. Local minimum

c. Local maximum

d. Inconclusive

$f_x = 2x, f_y = -2y \Rightarrow (0, 0)$. $f_{xx} = 2, f_{yy} = -2, f_{xy} = 0$. $D = (2)(-2) - 0^2 = -4 < 0$. Therefore $(0, 0)$ is a saddle point.

8 What does the Second Partials Test conclude when $D = 0$?

a. **The test is inconclusive; higher-order analysis or algebraic examination is needed ✓**

b. The point is definitely a saddle point

c. The point is an absolute extremum

d. The function is not differentiable

When $D = 0$, the test yields no information; the critical point could be a local maximum, local minimum, or saddle point.

9 Classify $(2, -3)$ for $f(x, y) = x^2 + y^2 - 4x + 6y + 10$.

a. **Local minimum ✓**

b. Local maximum

c. Saddle point

d. Inconclusive

$f_{xx} = 2, f_{yy} = 2, f_{xy} = 0$. $D = (2)(2) - 0 = 4 > 0$. Since $f_{xx} = 2 > 0$, it is a local minimum.

Section 4: Lagrange Multipliers for Constrained Optimization

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10 What vector equation governs constrained optimization of $f(x, y)$ subject to constraint $g(x, y) = k$?

a. $\nabla f = \lambda \nabla g$ **(and** $g(x, y) = k$**)** ✓

b. $\nabla f \cdot \nabla g = 0$

c. $\nabla f + \nabla g = \mathbf{0}$

d. $\nabla f = \mathbf{0}$

At a constrained extremum, the level curve of f is tangent to the constraint curve $g = k$, meaning their gradient vectors must be parallel: $\nabla f = \lambda \nabla g$.

11 Maximize $f(x, y) = xy$ subject to the constraint $x + y = 10$ for $x, y > 0$.

a. **Maximum is** 25 **at** $(5, 5)$ ✓

b. Maximum is 24 at $(4, 6)$

c. Maximum is 21 at $(3, 7)$

d. Maximum is 16 at $(2, 8)$

$\nabla f = [y, x]$ and $\nabla g = [1, 1]$. $[y, x] = \lambda[1, 1] \Rightarrow y = \lambda$ and $x = \lambda \Rightarrow x = y$. Using $x + y = 10 \Rightarrow 2x = 10 \Rightarrow x = 5, y = 5$. Maximum $f(5, 5) = 25$.

12 What does the Lagrange multiplier λ represent physically/economically in constrained optimization?

a. **The sensitivity / shadow price: the rate of change of the optimal value with respect to the constraint constant** k **(**$\frac{df^*}{dk}$**)** ✓

b. The curvature of the constraint surface

c. The distance from the origin to the optimal point

d. The second derivative along the constraint

The multiplier $\lambda = \frac{\partial f^*}{\partial k}$ represents the marginal change in the optimized objective value per unit increase in the constraint resource k.

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